Cosine Effect and the Velocity Measurement

To what extent does the angle of incidence between a moving object and the motion sensor affect the measured velocity?

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1. Introduction

Advanced Driver-Assistance Systems (ADAS) increasingly rely on motion detection technology to determine the position and velocity of surrounding objects. While these devices offer significant potential for enhancing vehicle safety, concerns persist regarding the accuracy and precision of their measurements. One potential source of error is the 'cosine error,' a systematic deviation arising from the velocity equation when the motion of the object being detected is not radial to the device.

This essay will examine the impact of cosine error on the accuracy of velocity readings from ultrasound detectors. Specifically, the research question is: To what extent does the angle of incidence between a moving object and the motion sensor affect the measured velocity?

2. Background Information

In many cases, the kinematics formula is used to calculate the velocity of objects detected by the motion sensors. When an ultrasound wave is projected onto an object, the object reflects the wave in the reverse direction. However, the angle of the detector relative to the motion of the object (angle of incidence) would affect the readings of its velocity. Therefore, this essay will further investigate the relationship between the angle of incidence of the ultrasound wave and its readings.

As mentioned above, the device has shortcomings when the beam of the ultrasonic sensor is not parallel to the direction of travel of the object detected. The angle of

incidence of the ultrasound wave affects how the waves travel, impacting the accuracy of the motion sensor readings in the experiment. This is known as the Cosine Error or Cosine Effect. To further understand the Cosine Effect, we must first briefly explain/review the formula of kinematics.

a. Velocity

Displacement is a measure of an object's change in position, including both the distance and the direction of that change. The formula is derived from the displacement formula of Topics A and C of the IB Physics SL/AHL curriculum.

Formula I: $d(t) = v_{sound} \cdot t \div 2$

$d(t) = Distance\ (Displacement\ from\ the\ point\ of\ measurement)$

$v_{sound} = Speed\ of\ ultrasound \approx 340\ ms^{-1}$

$t = Time\ taken\ to\ receive\ the\ rebounded\ wave$

Velocity, on the other hand, describes how fast an object's distance changes over time. It is the rate of change of distance. Therefore, besides the formulas given in the IB Data Booklet, the velocity can also be obtained by taking the derivative of displacement for time.

Formula II: $v(t) = \frac{d}{dt}(x(t))$

$v(t) = Velocity\ of\ object$

For the setup (5. Materials a. Motion Detector 2) that this experiment utilizes, it calculates the velocity using the formulas above. The detector pings and

measures the distance to objects. Based on the measurements and the calculation based on the formulas, the device's software (5. Materials b. Vernier Logger Pro 3) attempts to plot a distance graph. However, there are issues with the graphing, which are explained below in 5. Materials b. Vernier Logger Pro 3.

b. Cosine Error

When the ultrasound beam is projected out to the object, the waves are reflected to the device in the exact opposite direction. The displacement detection and calculation are only measured in this specific "line" of the ultrasound beam. As a result, the displacement and velocity vector directions are the same as the beam. However, in many cases, the object being detected is NOT travelling in the same direction as the ultrasound waves. This implies that if the beam and the object are not travelling in the same direction, the velocity detected would only be in the vector direction of the ultrasound beam instead of the direction of the object. (*Technical shortcomings of Doppler Traffic Radar*)

Formula III: $v_{measured} = v_{actual} \cdot cos(\theta)$

$v_{measured} = Velocity\ calculated\ from\ displacement\ and\ time$

$V_{actual} = Actual\ velocity\ of\ the\ object$

$\theta = Angle\ between\ the\ ultrasound\ beam\ and\ the\ object's\ direction\ of\ motion$

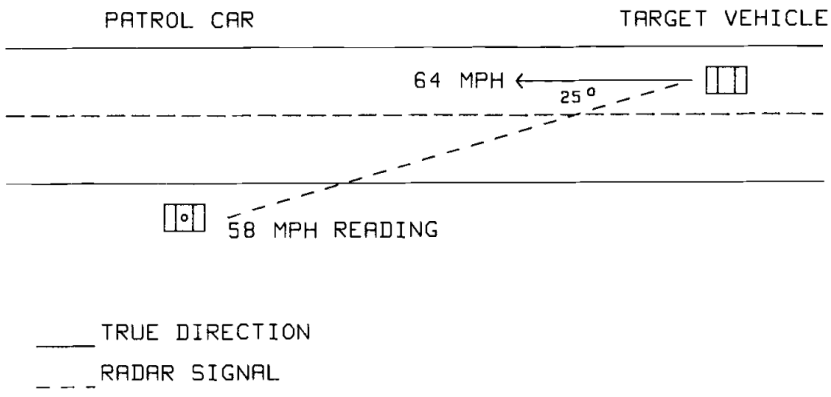


FIG. 3—*Because the target vehicle travels at a 25° angle with respect to the microwave beam, the true 103-kph (64-mph) speed is instead displayed as 93-kph (58-mph).*

Figure 1: Example diagram explaining the cosine error with an example. (*Technical shortcomings of Doppler Traffic Radar*)

3. Methodology

For this experiment, a moving cylinder is placed on the top of the LEGO train (5. Materials d. Train Set) on the track. The motion sensor is facing the track while placed a short distance from the track. During the trial, the cylinder will move on the track at a constant velocity, and the motion sensor will emit a beam against the object to determine its velocity. In each repetition of the experiment, the motion sensor would be facing a different section of the track. This is to adjust the angle of incidence of the ultrasound wave relative to the object's motion.

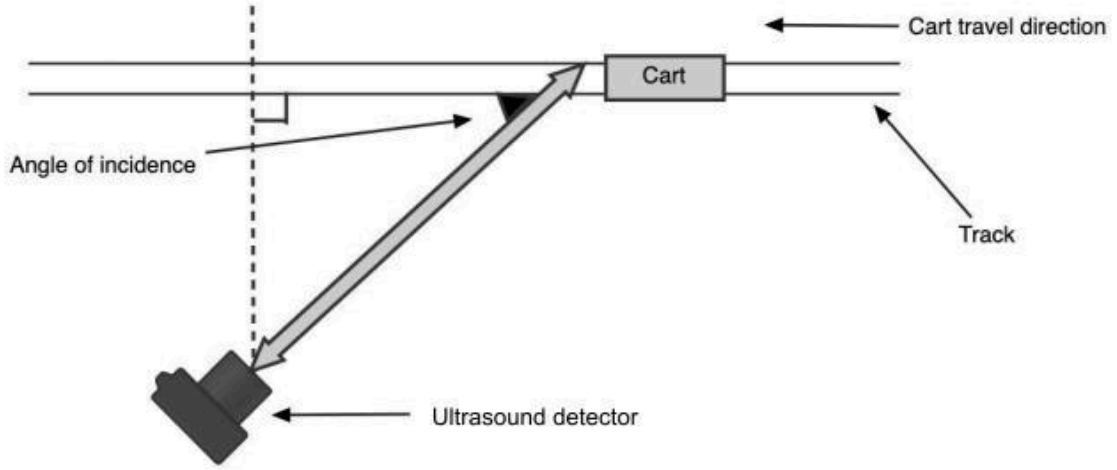


Figure 2, a diagram of how the data will be collected.

4. Variables

Type	Variable Name	Value / Description	Justification / Notes
Independent	Angle of incidence of the ultrasound beam	$30°, 25°, 20°, 15°, 10°, 5°, 0°$	Altered in fixed increments to investigate the angular dependence of the motion sensor.
Dependent	Measured speed via the cosine effect formula	Calculated from the recorded velocity	
Controlled	Velocity of the cart (object in motion)	Constant (maintained by motor LEGO control)	Ensures that speed variation is only due to angle, not object acceleration.
	Dimensions of the reflecting object	Height: approx. $10\ cm$; Diameter: approx. $20\ cm$	Kept constant to ensure a consistent reflection area and detection cross-section.

	Shape and material of the reflecting object	The cart was wrapped in aluminum foil	Ensures consistent ultrasound wave reflectivity across trials.
	Direction of the object's motion.	Linear motion along the track, toward the motor	Ensures motion is always directly approaching the ultrasound beam for a consistent Doppler shift.
	Motion sensor module specifications	Motion Sensor 2 by Vernier Dynamics	Fixed motion sensor ensures consistency in emitted wave frequency and sensor response.
	Protractor	Uncertainty: $\pm 1°$	Printed protractor paper
	Motion sensor measuring frequency	40 measurements per second.	Modified in the settings of the Logger Pro

5. Materials

a. HB100 Doppler Radar and Arduino (Scrapped)

Originally, this experiment planned on using the HB100 Doppler radar sensor to detect frequency shifts in reflected microwaves caused by the Doppler effect and thereby determine the velocity of a moving object along the sensor's line of sight (*HB100 microwave module manual*). Because the motion sensor's output signal is weak, an LM358 module was used to amplify the signal. Calibration of the amplifier was necessary to optimize gain and reduce distortion in the signal (*LM358 100 gain operational*

amplifier module). However, once the system was tested, the refresh speed was too slow, and the data obtained was neither precise nor accurate. After consultation with my supervisor, there was no cheap replacement for the Doppler radar. I decided to use the Vernier Motion Sensor 2 instead and investigate the cosine effect without the Doppler formula.

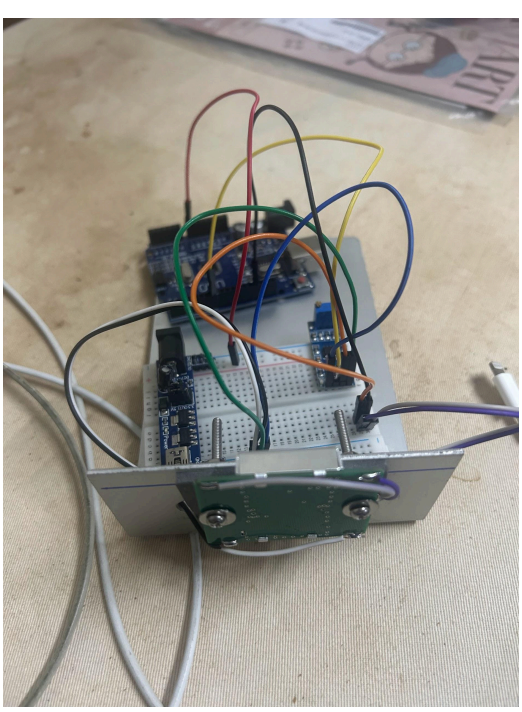




Figure 3 on the left, the inaccurate radar controlled using an Arduino UNO; Figure 4 on the right, the motion sensor replacing the Arduino radar.

b. Vernier Motion Detector 2

The Vernier Motion Detector 2 is an ultrasonic motion sensor that measures the distance of objects in real time. It operates by emitting ultrasonic pulses and detecting the time it takes for the echoes to return from a target, allowing for precise tracking of an object's movement. It

connects seamlessly with data collection software such as Logger Pro (explained below), enabling detailed analysis through real-time graphs and numerical data output.

An advantage of an ultrasonic sensor over Doppler radars in this experiment lies in its core output principle. Ultrasonic sensors measure the displacement (position) of the object by timing the sound pulse's reflection. The velocity is then calculated using the derivative of the displacement. Doppler radars, by contrast, measure a frequency shift that must be converted into velocity through complex calculations. This implies that the distance of the object from the motion sensor is shown. Although this distance is not used directly, it can serve as a rough verification mechanism to ensure that the sensor is roughly detecting the correct object and not background noise.

However, this motion sensor has several limitations, specifically with the detection zone. First, it detects objects from a range of $0.15m$ to $6m$. Any measurement above or below the range should be neglected. Second, the ping of the device can be illustrated as a cone of $15°$, resulting in measurements outside the cone that will not be accurate. Moreover, the detector can ping at an interval of most frequent every 0.05 seconds. The Vernier Logger Pro 3 (explained below) does not allow the interval to be set lower. Yet it is unclear whether the limit is a physical constraint or a software lock. Additionally, the device detects the main objects within the range and scope of the ping. Therefore, the detector measures the

distance of any fixed object within the detection zone (refer to below). The detector measures the train only when the train is in the detection zone. As a result, the distance will be measured regardless of the object.

Besides, the sensor does not show the device uncertainties. The uncertainty is not specified in the packaging or the official online manual by Vernier Systems. Oftentimes in such cases, the uncertainty is the smallest digit of measurement of the sensor. However, it is impossible to obtain that either (see b. Vernier Logger Pro 3).

The following are the summarized constraints of the Vernier Motion Detector 2:

i. Range: $0.15 - 6m$
ii. Cone of $15°$
iii. Most frequent every $0.05s$
iv. Only focuses on the “biggest object”
v. Unclear about uncertainties

c. Vernier Logger Pro 3

Vernier Logger Pro is the official software program for analyzing data collected from the Vernier Motion Detector 2. Based on the distances collected by the sensor, the software plots the data onto a table and a graph. The detector only measures distances and times. Then, the software uses these distances and times to calculate velocity and

acceleration. It doesn't just calculate the slope between two consecutive times; it appears to smooth out the velocity and acceleration graphs, likely by using a higher-degree polynomial (with higher-order derivatives), and that is why the software measures unreal speeds even after several pings have caught the train. Based on the gradient of the lines of the distance graph, the velocity of the objects is calculated. However, since the software only draws “perfect straight lines”, it filters out the small changes in the distance when drawing the line. These small changes could likely be due to the train’s movement. Thus, the distance and velocity graphs are not accurate representations of the train's movement.

Moreover, the Motion Detector 2 only focuses on the “closest object”. When the train is not in the zone of detection, the software shows the sensor detecting movement in the background ($> 1\ m$); yet when the train enters the zone of detection, the sensor focuses on the train ($< 1\ m$). This jump in distance over a short period of time results in unnaturally large velocities. Therefore, the velocity graph is completely disregarded. The method used to calculate the velocity is explained further in the Procedure.

Apart from the actual values, the decimal places of the values in the table are also inconsistent. As mentioned above, the uncertainty of the sensor is the smallest digit of the values recorded in the software. Conversely, the

Vernier Data Logger Pro 3’s tables are set up in the spreadsheet style, rounding up values, making it impossible to find the uncertainty of the sensor.

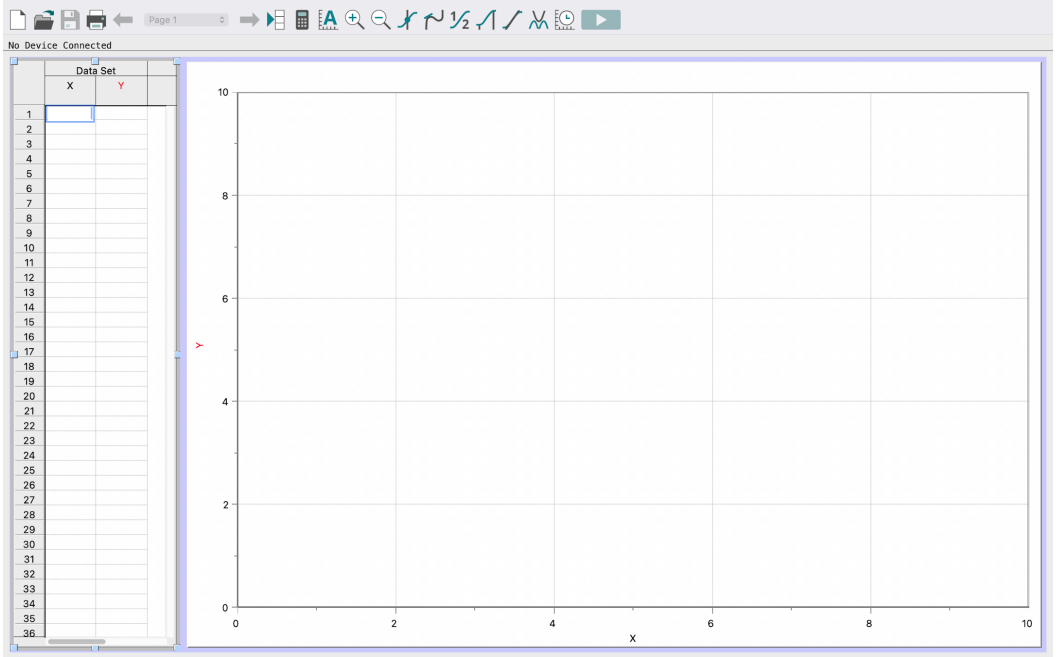


Figure 5, Data logger Pro

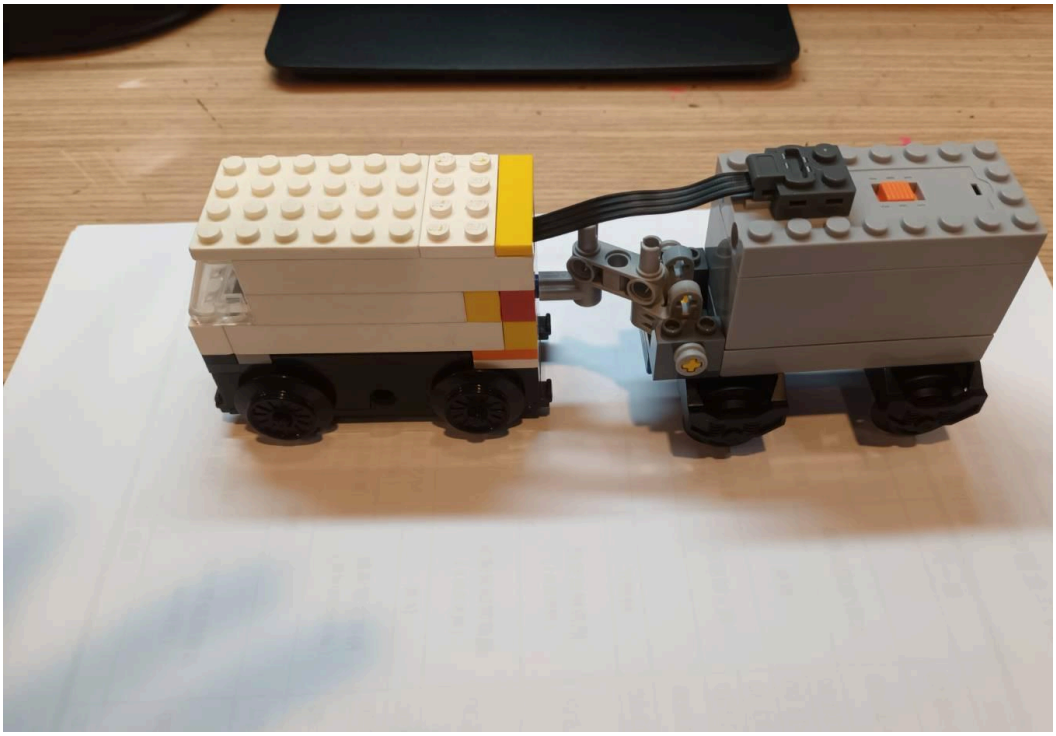
d. Train Set

To ensure constant linear velocity of the cart, a LEGO microcontroller is integrated into the system. The motor is controlled via a LEGO motor module, which supplies stable voltage and current to the motor.

As mentioned above, the sensor detects the shortest distance regardless of the object. There is a likelihood that the sensor measures different parts of the train. Since the train has approximately rectangular dimensions, the measurement of the distance would remain constant as the sensor measures the distance of different parts of the train. Consequently, the

velocity is measured to be 0 even though the train is moving. During our first time during the experiment, we were not aware of the issue. It has resulted in the data collected being useless.

To mitigate this problem, cotton balls are taped onto the sides of the train, and a foil-covered cylinder taken from a pen is taped on the train. As a result, the motion sensor only detects the cylinder, greatly reducing the cross-sectional area, which improves the accuracy.





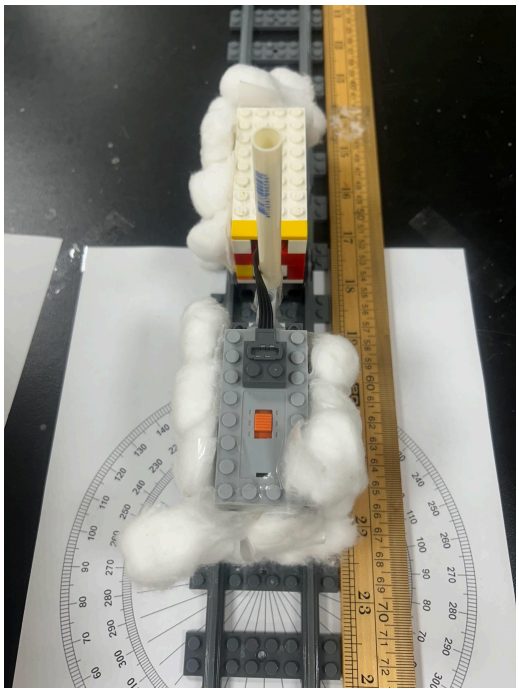


Figure 6 on top, LEGO train; Figure 7 on the bottom left, the cotton balls; Figure 8, train with the cotton balls and the stick (without foil wrap)

e. Large Screen

Since the Vernier Motion Detector 2 can detect objects extremely far ($< 3m$), a screen is needed to block unwanted motion behind the train. In this case, a large umbrella is used. This ensures that the motion sensor only picks up the train and not other background disturbances, which would lead to inaccurate measurements.



Figure 9, large screen.

6. Procedure

a. Setup

i. The experimental setup was constructed to facilitate controlled linear motion of a cart while simultaneously recording displacement shifts via an ultrasound motion sensor. During the initial construction phase, a LEGO-powered cart is covered with foil. The train cart was then placed on a set of straight LEGO tracks.

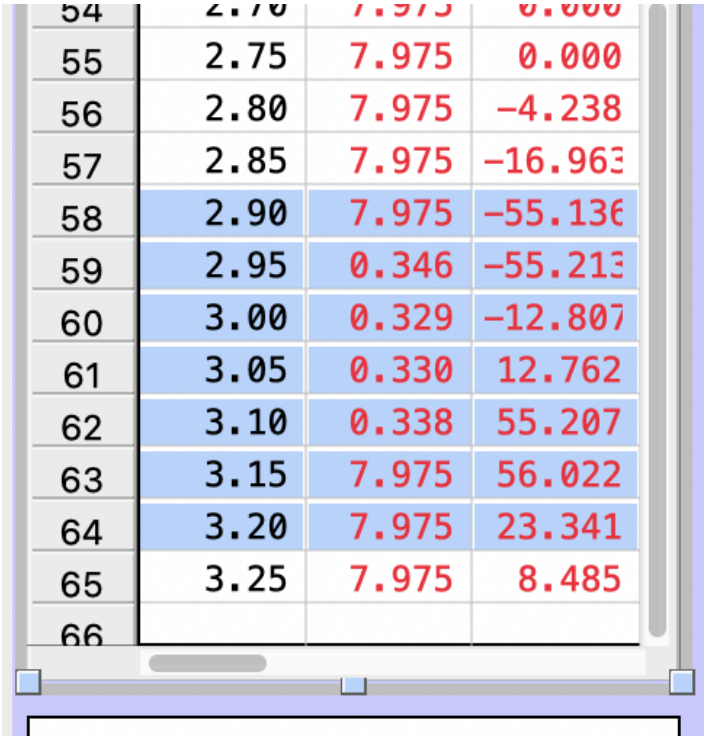
ii. To collect data, the Vernier Motion Sensor 2 module was initially positioned at an angle of incidence of $0°$ relative to the track. The train ran at constant velocity. The sensor was used to record the cart's velocity (Refer to Data Collection on how data was recorded). Once the cart exited the motion sensor's effective detection zone or the velocity readings stabilized, the motor was halted. This process was repeated for four trials at each angle to ensure measurement consistency.

iii. Subsequently, the angle of incidence of the motion sensor was increased in $5°$ increments ($0°, 5°, 10°, 15°, 20°, 25°, 30°$), with the same procedure followed at each angle. Data from the Vernier Motion Sensor 2 was recorded for all trials to enable a detailed comparison and analysis of how the velocity calculated varies with the angle of observation.

b. Data Collection.

Data was collected using a Vernier Motion Sensor 2 interfaced with Vernier Logger Pro software. As mentioned above, the software provides instantaneous velocity readings by calculating the displacement over very short time intervals. However, some irrelevant data points should be disregarded.

To ensure the validity of the velocity measurements, only data points recorded when the cart was clearly within the sensor's detection zone were considered. The sensor detects the distance and calculates the velocity regardless of what the object is. Therefore, to collect the data, the change in distance over time (Formula II above) when the sensor is detecting the train (distance less than 1 meter)



55	2.75	7.975	0.000
56	2.80	7.975	−4.238
57	2.85	7.975	−16.963
58	2.90	7.975	−55.136
59	2.95	0.346	−55.213
60	3.00	0.329	−12.807
61	3.05	0.330	12.762
62	3.10	0.338	55.207
63	3.15	7.975	56.022
64	3.20	7.975	23.341
65	3.25	7.975	8.485
66			

Figure 10, only considering the data when the train is close in the frame of the detector (roughly 0.3 m, second column). Note that the leftmost column is the time (s); the center column is the distance (m); and the right column is the velocity.

Therefore, the velocity data must be calculated manually from the distance. To obtain the velocity of the train, only the portion when the distance is close ($< \ 1m$) is focused on. Therefore, the velocity is calculated by taking the distance change over the change in time. For example, using the data from Figure 10, the distance difference is $0.338 - 0.329 = 0.009$, and the time difference is $3.10 - 3.00 = 0.10$. Thus, the calculated velocity is $0.009 \div 0.1 = 0.09ms^{-1}$.

While data will be collected from angles $0°$ to $30°$, measurements at $0°$ will not be processed with the other angles. Instead, the observed velocity at $0°$ is assumed to be the real velocity of the train, which will be used to

compare to the calculated real velocity from the slope of the line of the graph. This is explained further below in the processing section.

7. Recorded Data

Trial	Angle (degrees)	Observed Velocity (m/s)				
		1	2	3	4	5
a	0	1.2429	1.2559	1.27	1.3366	1.189781
b	5	1.306144	0.910911	1.39258	1.31026	1.280248
c	10	1.444716	1.469755	1.39601	1.482903	1.408587
d	15	1.42884	1.3809	1.4334	1.4835	1.3734
e	20	1.3771	1.4097	1.3984	1.4234	1.3549
f	25	1.324	1.2725	1.2293	1.3686	1.2262
g	30	1.3943	1.3871	1.4036	1.395	1.2808

8. Processing

All example calculations in the processing will use trial a1 or group a collectively as example data.

a. Actual Values

i. Remove outliers from the raw data using IQR.

Example Calculations:

$$IQR = Q_3 - Q_1 = 1.3033 - 1.216340625 = 0.086959375$$

$$Outlier\ if\ >\ (Q_3 + 1.5 \cdot IQR)\ or\ <\ (Q_1 - 1.5 \cdot IQR)$$

Outliers: a4, a5, b2, b3

ii. Find the average values for each group (same angle).

Example calculations:

$$V_{avg} = \frac{V_1+V_2+V_3+V_4+V_5}{5} \approx 1.256266667$$

Trial	cos(θ)	Observed Velocity (m/s)
a	1	1.256266667
b	0.996195	1.303845359
c	0.984808	1.462614602
d	0.965926	1.470100458
e	0.939693	1.482080384
f	0.906308	1.416869653
g	0.866025	1.584433891

Note: According to Formula III, the observed velocity should increase when the cosine value decreases (angle of incidence increases).

b. Uncertainties

In this experiment, we only considered the uncertainties associated with the protractor paper and the Vernier Motion Detector 2, both of which contribute to the calculated velocity of the train.

i. Protractor Paper Uncertainty

The protractor paper with $1°$ increments, has an absolute uncertainty of $\pm 1°$. However, when calculating the percentage uncertainty, the following formula is used:

$$\%Unc = \frac{cos(Value+abs_unc)+cos(Value-abs_unc)}{2}$$

$$abs_unc = 1$$

$$Value = actual\ value\ of\ the\ angle$$

ii. Observed Velocity Uncertainty

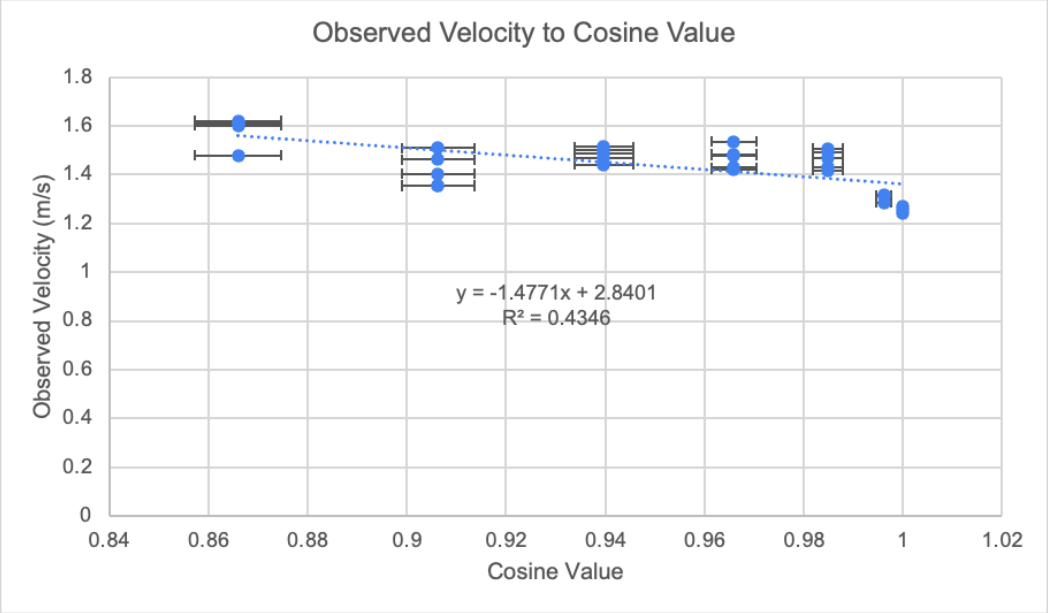
As mentioned in 5. Materials, the official manual for the Vernier Motion Detector 2 does not specify its uncertainty. Yet, based on the software, the sensor's uncertainty is minimal, which makes the uncertainty value unusable. Instead, in the graph of the average observed velocity and cosine value (Figure 12), the uncertainty of the velocity is half of the range of the values.

c. Graphing

Data from different angles, ranging from $5°$ to $30°$, is graphed. However, the observed velocity from $0°$ is intentionally left out. This is because data from $0°$ is used to compare with the actual velocity obtained from the gradient of the graph.

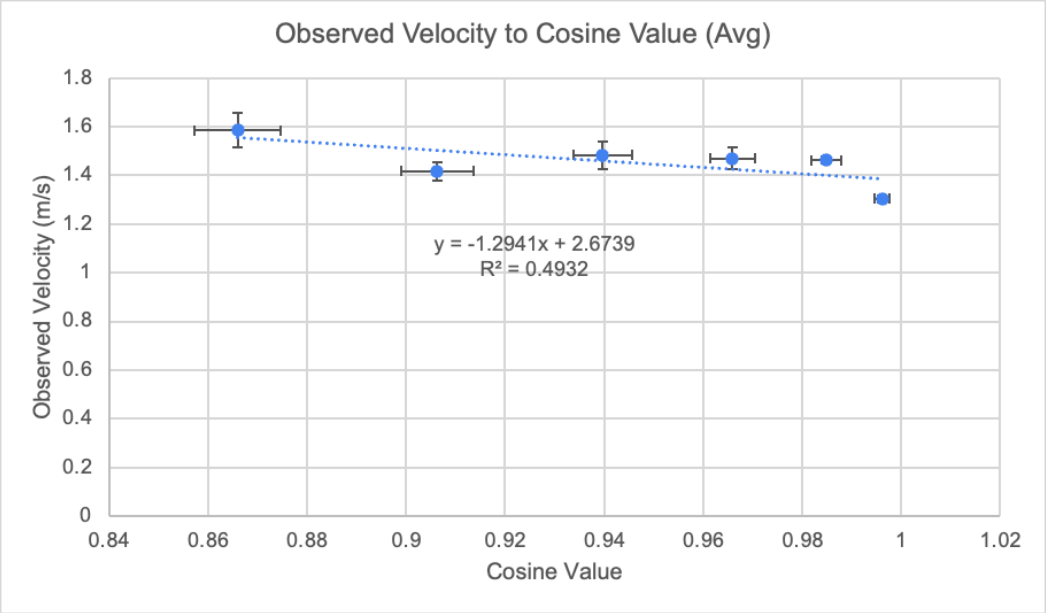
The observed velocity is on the y-axis, while the cosine value is on the x-axis. This is done to find the calculated actual velocity through the slope of the line of best fit. According to Formula III, $V_{actual} = \frac{V_{measured}}{cos(\theta)}$

i. Graph all the data with all angles except $0°$ (see paragraph above for a reason). Figure 11.



Note: the uncertainty of the Observed Velocity is too small to be visible.

ii. Graph the mean data with all angles except $0°$ (see paragraph above for a reason). Figure 12.



iii. Utilizing the graph to obtain the velocity of the train and the line strength.

To obtain the strength of the best fit, or the R^2 value, the simple method is to use the automatically calculated value on the graph. With the graph, the velocity can also be obtained by the gradient.

$$V_{Calculated} \approx -\ 1.29ms^{-1}$$

$$R^2 = 0.49$$

$$y_{intercept} = 2.6739$$

iv. Calculate the percentage error of the values.

$$\%Error = \frac{V_{Actual} - V_{Calculated}}{V_{Actual}} \cdot 100\% \approx 3.01\% \approx 3\%$$

$$V_{Actual} = Observed\ speed\ of\ train\ at\ 0°$$

9. Analysis

The calculated velocity is negative since the gradient is negative. This indicates that when the cosine value increases, the observed velocity decreases. Since the cosine value between $0°$ and $90°$ (able to be observed in nature) decreases as the angle increases, the observed velocity decreases when the angle of incidence decreases. Because the actual velocity of the train is considered to be constant (4. Variables), the trend does not support Formula III in the Background section. Using the value of the gradient on the graph and the mean velocity at $0°$, the percentage error is at 3%, which is low for scientific experiments, further substantiating a digression from Formula III.

However, since the deduction et al is based on the line of best fit in the graphs, it is important to consider the strength of this correlation. In this case, the R^2 value is used. R^2 values are between 0 and 1, with 1 being perfectly correlated and 0 indicating no correlation. In this experiment, $R^2 \approx 0.49$ there is a moderate to low correlation. There is noise in the data points, making the data numerically scattered. Pragmatically, the low correlation and the noise could be a result of multiple factors, most likely due to systematic and random errors.

a. Systematic Error

A significant source of systematic error in this investigation stemmed from the unverified assumption of constant velocity convergence. In theory, the train's

velocity should asymptotically approach a terminal value after traversing a sufficient distance under constant acceleration from rest. However, given the limited length of the experimental track, it was not definitively established whether the train achieved this terminal velocity within the measurement range, or indeed if velocity convergence occurred at all. In this experiment, all velocities were recorded at $\frac{3}{4}$ of the track. However, there is still potential for increasing velocities at the point of measurement across different trials, introducing a systematic deviation in the dependent variable.

To mitigate this systematic error in future experiments, the track length should be increased to approximately 3 to 4 meters. This extension would allow for a greater displacement, thereby increasing the likelihood of the train attaining its terminal velocity before measurement. Furthermore, the motion sensor should be set to initiate velocity measurements exclusively when the train is nearing the end of the extended track. This ensures that data is collected only during the phase where the train's velocity is most likely to be constant, thus enhancing the validity of the measurements.

Another substantial systematic error was attributed to the limitations of the motion sensor and its accompanying software. The sensor exhibited insufficient precision and lacked documented uncertainty specifications. Furthermore, the

raw data collected by the motion sensor contained significant noise. While the Vernier Logger Pro software employed filtering algorithms to remove these erroneous values, the efficacy of this process was questionable, as evidenced by the persistence of numerous unfiltered outliers. The proprietary and undisclosed nature of this filtering process further compromised the precision of the collected data.

To address these issues, future iterations of this experiment should utilize motion sensor systems with superior precision and resolution, paired with more robust data acquisition and analysis software. Such systems should ideally provide explicit information regarding their measurement uncertainty and the methodologies employed for noise reduction. The availability of this information would not only enhance the precision of the measurements but also significantly increase the overall rigor and transparency of the experiment.

One other systematic error is that the sensor could still be detecting the different parts of the train. Despite the efforts to reduce reflection on the rectangular part with the cotton, there is still the possibility that the same energy was reflected and that multiple sections of the train were detected, affecting the observed velocity. To resolve this problem, better equipment can be used to absorb the waves from the ultrasound, controlling the detection cross-section.

b. Random Error

Random fluctuations in velocity measurements were observed across multiple trials, leading to inconsistencies in the recorded data. These variations are characteristic of random errors inherent in experimental procedures. To minimize the impact of this random error on the reliability of the results, five repeated trials were conducted for each experimental condition. While this approach helps to average out random variations, the precision of the measured velocities could be further improved. Therefore, future investigations could implement an increased number of trials (e.g., 10 or more) to further reduce the statistical significance of random fluctuations and enhance the robustness of the data.

10. Conclusion

This investigation does show that the measured velocity from an ultrasonic motion sensor decreases with decreasing angle of incidence (cosine value increases), not aligning with the theoretical cosine error model. By comparing measured velocities across multiple angles and averaging trial results, the data do not support the relationship $v_{measured} = v_{actual} \cdot cos(\theta)$.

The overall pattern of results did not validate the central hypothesis. The relatively low percentage error between predicted and observed values indicates that the cosine

effect remains a dominant source of error in non-radial velocity measurements using ultrasonic sensors.

These findings emphasize the importance of sensor alignment in applications like motion tracking and automotive systems. Even small deviations from head-on alignment can introduce measurable inaccuracies, reinforcing the need for optimized sensor placement in real-world scenarios.

Completing this experiment marked a significant journey of academic and personal growth. The initial failure with the Doppler Radar forced a critical pivot, teaching me experimental resilience and resourcefulness through real-world technical challenges. Furthermore, manually calculating velocity from raw data and disregarding incorrect automated outputs has improved my skills in data integrity and critical evaluation. This process matured my approach to complex, independent scientific inquiry, preparing me well for future academic pursuits.

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